Insertion Sort – Code and Output

C Program

```
#include <stdio.h>

// Function to print the array elements
void printArray(int arr[], int size) {
    for (int i = 0; i < size; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}

// Function to implement insertion sort
void insertionSort(int arr[], int size) {
    int key, j;

    // Outer loop starts from the second element (index 1)
    for (int i = 1; i < size; i++) {
        key = arr[i];
        j = i - 1;

        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j = j - 1;
        }

        arr[j + 1] = key;

        printArray(arr, size);
    }
}

int main() {
    int data[] = {12, 11, 13, 5, 6};

    int size = sizeof(data) / sizeof(data[0]);

    printf("Original array:\n");
    printArray(data, size);

    insertionSort(data, size);

    printf("Sorted array in ascending order:\n");
    printArray(data, size);

    return 0;
}
```

Output

```
OUTPUT:

Original array:
12 11 13 5 6

11 12 13 5 6
11 12 13 5 6
5 11 12 13 6
5 6 11 12 13

Sorted array in ascending order:
5 6 11 12 13
```